AI-Enabled Material Submittal & Compliance Review System

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Version: 1.0 (MVP)
Document Type: Formal Software Requirements Specification
Prepared For: Construction Consultants, Contractors & Suppliers

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1. INTRODUCTION

1.1 Purpose

This document defines the functional and non-functional requirements for a **multi-tenant, AI-assisted Material Submittal Management System**.
The system is intended to **digitize, standardize, and partially automate** the material submittal, compliance checking, and review workflow commonly used in construction consultancy projects.

The SRS serves as a reference for:

- Product design and development
- Client and consultant alignment
- Costing and implementation planning

1.2 Scope

The system will:

- Replace email-based and hard-copy material submittals
- Provide structured digital material submission forms
- Perform **AI-based preliminary compliance screening**
- Support detailed **engineer review, comments, approval, and resubmission**
- Maintain full audit trails and revision history

The **final approval authority remains with human engineers**.

1.3 Definitions & Abbreviations

Term	Meaning
SRS	Software Requirements Specification
AI	Artificial Intelligence
OCR	Optical Character Recognition
MVP	Minimum Viable Product
PQ	Pre-Qualification
BMS	Building Management System

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2. OVERALL SYSTEM DESCRIPTION

2.1 Product Perspective

The platform is a **cloud-based, web-based application** built using a **multi-tenant architecture**, where each consultancy operates in complete data isolation.

It replaces:

- Email submissions
- Manual document tracking
- Unstructured engineer comments

with a **centralized, auditable, and intelligent system**.

2.2 User Classes & Roles

Role	Description
Super Admin	Platform configuration, tenant management
Consultant Admin	Project setup, workflow control
Consultant Engineer	Technical review and approval
Contractor	Material submittal creation
Supplier	Upload manufacturer documents

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3. FUNCTIONAL REQUIREMENTS

3.1 Tenant & Project Management

FR-01

The system shall support multiple tenants with complete data isolation.

FR-02
Each tenant shall be able to create and manage multiple projects.

FR-03
Each project shall support its own:

- Material Submittal Index
- Specification documents
- Approval workflows

3.2 Material Submittal Creation

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The system shall allow contractors/suppliers to create **individual material submittals** per project.

FR-05
Each material submittal shall support **multiple document uploads** mapped to a predefined Material Submittal Index.

FR-06
The system shall enforce **mandatory document checks** before allowing submission.

3.3 Structured Material Data Capture

Each material submittal **must capture structured technical data**, including:

Material Description

- Material type
- Grade / class
- Finish / coating
- Size / thickness / rating

Manufacturer Details

- Manufacturer name
- Product model / code
- Country of origin
- Manufacturer authorization (if applicable)

Standards & Specifications

- Applicable ASTM / BS / EN / ISO standards
- Project specification section references
- Authority / regulatory requirements

3.4 Document Types (Configurable Index)

The system shall support documents including but not limited to:

- Company profile
- ISO & quality certificates
- Compliance statements
- Datasheets (highlighted)
- Test reports
- Calculations
- Authorities approvals
- Warranty letters
- Country of origin certificates
- Drawings, schematics, SLDs
- Sample photos

3.5 AI-Assisted Pre-Screening (MVP)

FR-07

The system shall use AI to perform **initial document screening** before engineer review.

AI capabilities in MVP:

- OCR on scanned PDFs and images
- Document classification (certificate, datasheet, test report, etc.)
- Completeness checks against the required index
- Detection of missing, expired, or unclear certificates
- Identification of missing manufacturer or origin details

AI output:

- Likely compliant
- Missing / unclear
- Non-compliant (with reason)

AI results are advisory and do not auto-approve materials.

3.6 Compliance Matrix Generation

FR-08
The system shall auto-generate a **Compliance Matrix** mapping:

- Project specification clauses
- Submitted documents
- Compliance status

This matrix shall be visible to engineers during review.

3.7 Consultant Review Workflow

FR-09
Consultant engineers shall review:

- Submitted documents
- AI screening results
- Compliance matrix

FR-10
Engineers shall be able to:

- Approve
- Approve with comments
- Revise & resubmit
- Reject

FR-11
All comments shall be:

- Structured
- Linked to specific documents
- Saved with revision history

3.8 Revision & Resubmission Control

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The system shall support **multiple revisions per material submittal**.

FR-13
Each revision shall retain:

- Previous comments
- Previous documents
- Approval status history

3.9 Audit Trail & Traceability

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The system shall maintain a complete audit trail of:

- Submissions
- AI screening results
- Engineer actions
- Dates, users, and comments

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4. NON-FUNCTIONAL REQUIREMENTS

4.1 Security

- Role-based access control
- Secure authentication
- Encrypted document storage
- Tenant-level data isolation

4.2 Performance

- Support large PDF uploads
- Handle concurrent reviews
- AI screening performed asynchronously

4.3 Scalability

- Support thousands of materials per project
- Horizontal scaling for AI processing
- Long-term document retention

4.4 Compliance & Governance

- ISO 27001-aligned practices
- GDPR-ready data handling
- Contractual audit readiness

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5. AI DESIGN – MVP & FUTURE ROADMAP

5.1 MVP AI Capabilities

- OCR & document understanding
- Rule-based compliance checks
- Missing document detection
- Compliance summary generation

5.2 Future AI Enhancements (Post-MVP)

- Specification reasoning agents
- Clause-by-clause compliance validation
- Auto-draft engineer comments
- Cross-project learning of approvals
- Risk scoring of materials and suppliers
- Authority-specific compliance checks

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6. CONSTRAINTS & ASSUMPTIONS

- AI shall not replace final engineering judgment
- Final approval authority rests with consultant engineers
- System recommendations must be explainable

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7. SUCCESS METRICS

- Reduction in average review time
- Higher first-pass approval rate
- Fewer revise & resubmit cycles
- Improved engineer productivity
- Better compliance traceability